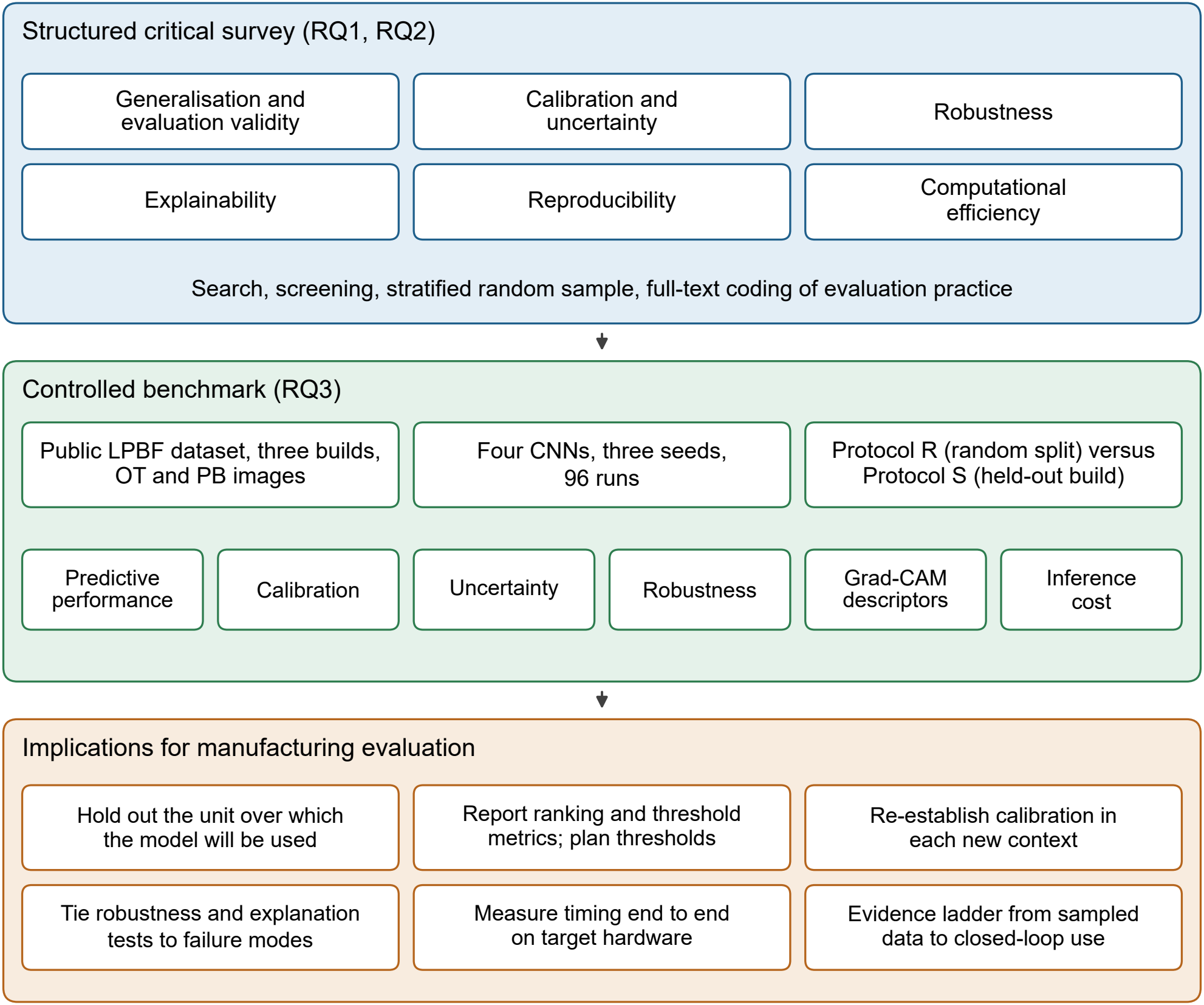
Structured critical survey (RQ1, RQ2)
Generalisation and evaluation validity
Calibration and uncertainty
Robustness
Explainability
Reproducibility
Computational efficiency
Search, screening, stratified random sample, full-text coding of evaluation practice
Controlled benchmark (RQ3)
Public LPBF dataset, three builds, OT and PB images
Four CNNs, three seeds, 96 runs
Protocol R (random split) versus Protocol S (held-out build)
Predictive performance
Calibration
Uncertainty
Robustness
Grad-CAM descriptors
Inference cost
Implications for manufacturing evaluation
Hold out the unit over which the model will be used
Report ranking and threshold metrics; plan thresholds
Re-establish calibration in each new context
Tie robustness and explanation tests to failure modes
Measure timing end to end on target hardware
Evidence ladder from sampled data to closed-loop use